What is this empty space within which particles move? Is it something, or is it nothing? If it is nothing, it doesn't exist, and we can do without it. If it is something, can it be true that its only property is to be there, *doing* nothing?

Since antiquity, the idea of empty space, halfway between a thing and a non-thing, had troubled thinkers. Democritus himself, who had placed empty space at the basis of his world where atoms course, certainly wasn't crystal clear on the issue: he wrote that empty space is something 'between being and non-being': 'Democritus postulated the full and the empty, calling one "Being", and the other "Non-Being",' says Simplicius.[1] Atoms are being. Space is non-being – a 'non-being' that, nevertheless, exists. It is difficult to be more obscure than this.

Newton, who resuscitated the Democritean idea of space, had tried to patch things up by arguing that space was God's *sensorium*. No one has ever understood what Newton meant by 'God's sensorium', perhaps not even Newton himself. Certainly, Einstein, who gave little credit to the idea of a God (with or without a sensorium), except as a playful rhetorical device, found Newton's explanation of the nature of space utterly unconvincing.

Newton struggled considerably to overcome the scientists' and philosophers' resistance to his reviving the Democritean concept of space; at first nobody took him seriously. Only the extraordinary efficacy of his equations, which turned out to predict always the correct outcome, ended up silencing criticism. But doubts concerning the plausibility of the Newtonian concept of space persisted, and Einstein, who read philosophers, was well aware of them. Ernst Mach, whose influence Einstein readily acknowledged, was the philosopher who highlighted the conceptual difficulties of the Newtonian idea of space – the same Mach who did not believe in the existence of atoms. (A good example, incidentally, of how the same person can be short-sighted in one respect and far-seeing in another.)

Thus, Einstein addresses not one but two problems. First, how can we describe the gravitational field? Second, what is Newton's space?

And it's here that Einstein's extraordinary stroke of genius occurs, one of the greatest flights in the history of human thinking: what if the gravitational field turned out actually to *be* Newton's mysterious space? What if Newton's space was nothing more than the gravitational field? This extremely simple, beautiful, brilliant idea is the theory of general relativity.